

Network representation

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Solution of the practice quiz

1. Here the cardinality of the set of nodes $\mathcal{N}$ is $|\mathcal{N}| = 6$. Then, considering two nodes $i, j \in \mathcal{N}$ and the set of arcs $\mathcal{A}$, the adjacency matrix $A \in \mathbb{R}^{6 \times 6}$ is defined as

$$A(i, j) = \begin{cases} 1 & \text{if } (i, j) \in \mathcal{A} \\ 0 & \text{otherwise.} \end{cases}$$

If we consider the natural numbering order from v_1 to v_6 , we obtain:

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

2. If we want to store the values of the flows, we can do several things. For example, we can use the same structure:

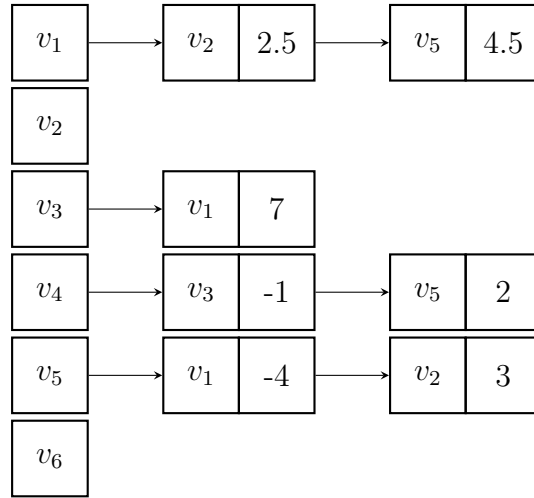
$$F = \begin{pmatrix} 0 & 2.5 & 0 & 0 & 4.5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 7 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 2 & 0 \\ -4 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The disadvantage is that we store a lot of zero values. As suggested in the video, we can also adopt a row-by-row numbering for the arcs, and store the flows into a vector

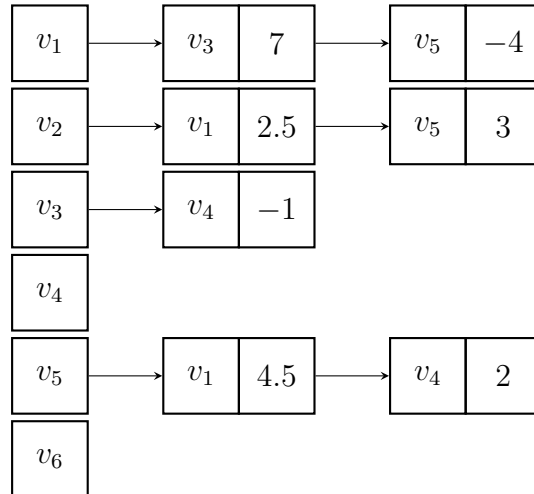
$$F = (2.5, 4.5, 7, -1, 2, -4, 3).$$

3. The adjacency list corresponding to node $i \in \mathcal{N}$ is a list of length equal to its outdegree d_i^+ (recall that the outdegree of node i is the number of arcs of the form $(i, j) \in \mathcal{A}$). Each element of the list corresponds to an arc going out of i .

For each arc, we also store the value of the flow. We obtain:



4. The adjacency lists for predecessor would be as follows:



For instance, node v_1 has two predecessors: v_3 and v_5 . The arc (v_3, v_1) carries a flow of 7, and the arc (v_5, v_1) carries a flow of -4 . Therefore, the adjacency list of node v_1 has two elements: node v_3 associated with the flow on arc (v_3, v_1) , that is 7, and node v_5 associated with the flow on arc (v_5, v_1) , that is -4 . Node v_4 has no predecessor. Therefore, its adjacency list is empty.